

Unit 2 Review*Real Numbers, Square Roots and Cube Roots — 41 questions*

HOW TO USE THIS. This review covers every learning target from all six lessons of Unit 2 — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Square Values and Square Roots

Lessons 1 and 2 · 9 questions

1. List every perfect square from 1 to 225.

2. Evaluate. $(-7)^2$

3. Evaluate. $\sqrt{196}$

4. Evaluate. $-\sqrt{169}$

5. Evaluate. $\sqrt{-49}$

6. A square patio covers 144 square feet. How long is each side?

7. Explain the difference between $\sqrt{81}$ and the solutions of $x^2 = 81$.

8. True or false: $\sqrt{225} = \pm 15$

9. Every part below is about $\sqrt{40}$.

a. Between which two consecutive integers does $\sqrt{40}$ lie, and which is it closer to?

b. Is $\sqrt{40}$ rational or irrational? How do you know?

c. Solve $x^2 = 40$. How many solutions does it have?

d. $\sqrt{40} \approx 6.3$. Explain why $\sqrt{40} \times \sqrt{40}$ is exactly 40 but 6.3×6.3 is not.

PART B. Cube Values and Cube Roots

Lesson 3 · 6 questions

10. List every perfect cube from -125 to 125 .

11. Evaluate. $(-4)^3$

12. Evaluate. $\sqrt[3]{125}$

13. Evaluate. $\sqrt[3]{-27}$

14. A cube holds 64 cubic centimeters. How long is each edge?

15. Why does $\sqrt[3]{-64}$ have a real value while $\sqrt{-64}$ does not?

PART C. Solving $x^2 = p$ and $x^3 = p$

Lessons 2 and 3 · 6 questions

16. Solve. $x^2 = 121$

17. Solve. $x^2 = 0$

18. Solve. $x^2 = -16$

19. Solve. $x^3 = 27$

20. Solve. $x^3 = -125$

21. Select all of the solutions of $x^2 = 100$.

☐ $x = -100$

☐ $x = -10$

☐ $x = 10$

☐ $x = 100$

PART D. Approximating Irrational Numbers

Lesson 4 · 6 questions

22. Between which two whole numbers does $\sqrt{30}$ lie? Which is it closer to?

23. Between which two whole numbers does $\sqrt[3]{70}$ lie? Which is it closer to?

24. Approximate $\sqrt{52}$ to the nearest tenth.

25. Approximate $\sqrt[3]{20}$ to the nearest tenth.

26. Approximate 2π to the nearest tenth.

27. Dev uses $\sqrt{11} \approx 3.3$ to approximate $4\sqrt{11}$ and gets 13.2. A calculator gives 13.3. What went wrong?

PART E. Rational or Irrational

Lessons 4 and 5 · 6 questions

28. Is $\sqrt{49}$ rational or irrational? Explain.

29. Is $\sqrt[3]{10}$ rational or irrational? Explain.

30. Is $0.\overline{4}$ rational or irrational? Explain.

31. Is -0.5π rational or irrational? Explain.

32. Is $\frac{\sqrt{144}}{4}$ rational or irrational? Explain.

33. Sort the six numbers into the table.

$$\sqrt{36}, \sqrt{37}, \frac{22}{7}, \pi, \sqrt[3]{8}, -\sqrt[3]{9}$$

Rational	Irrational
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PART F. Plotting, Ordering and Comparing

Lessons 5 and 6 · 8 questions

34. Order from least to greatest. $\sqrt{17}$, $-\sqrt[3]{30}$, 4.6 , -3.5 , $\frac{13}{4}$

35. Order from least to greatest. $-\sqrt{50}$, -2π , $\sqrt[3]{-20}$, $\frac{19}{4}$, $\sqrt{35}$

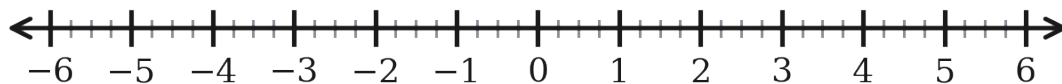
36. Compare using <, > or =. $\sqrt{26}$ 5.2

37. Compare using <, > or =. $-\sqrt[3]{65}$ -4

38. Compare using $<$, $>$ or $=$. 4π $\sqrt{150}$

39. Compare using $<$, $>$ or $=$. $\sqrt[3]{-8} \cdot 5$ $-\sqrt{100}$

40. Plot and label each value. $-\sqrt{20}$, $-\frac{5}{2}$, $\sqrt[3]{10}$, $\sqrt{28}$



41. Marcus orders $-\sqrt[3]{30}$, -3.2 , $-\sqrt{8}$ from least to greatest and writes $-\sqrt{8}$, -3.2 , $-\sqrt[3]{30}$. Is he right? If not, fix it.